GAYATRI VIDYA PARISHAD COLLEGE FOR DEGREE AND PG COURSES (A)

Department of Engineering & Technology

Scheme of Valuation

Course: Engineering Mathematics-II
Semester: II
MID-I Examination
Maximum Marks: 30

PART-A (3 × 2 = 6 Marks)

Q.1(a) Show that A is unitary if and only if $A^{-1} = A^H$. *(2 Marks)*

- Definition of unitary matrix – **0.5 Mark**
- Proof of necessity ($A^H A = I \Rightarrow A^{-1} = A^H$) – **0.75 Mark**
- Proof of sufficiency ($A^{-1} = A^H \Rightarrow A^H A = I$) – **0.75 Mark**

Q.1(b) Find the rank of the given matrix. *(2 Marks)*

- Correct row operations – **1 Mark**
- Determination of rank with justification – **1 Mark**

Q.1(c) Find the eigenvalues of the given matrix. *(2 Marks)*

- Formation of characteristic equation – **1 Mark**
- Correct eigenvalues – **1 Mark**

PART-B (Answer ONE question from each Unit)

Q.2(a) Find the rank of the matrix by reducing it to Echelon form. *(6 Marks)*

- Correct elementary row operations – **2 Marks**
- Reduction to echelon form – **2 Marks**
- Determination of rank with proper conclusion – **2 Marks**

Q.2(b) Find the rank of the matrix by reducing it to Normal form. *(6 Marks)*

- Elementary row and column operations – **2 Marks**
- Reduction to normal form – **2 Marks**
- Correct rank with justification – **2 Marks**

OR

Q.3(a) Show that the given system is consistent and solve it. *(6 Marks)*

- Formation of augmented matrix – **1 Mark**
- Row reduction / consistency check – **2 Marks**
- Identification of consistency – **1 Mark**
- Correct solution of variables – **2 Marks**

Q.3(b) Solve the system using Gauss Elimination Method. *(6 Marks)*

- Formation of augmented matrix – **1 Mark**
- Forward elimination – **2 Marks**
- Back substitution – **2 Marks**
- Final solution – **1 Mark**

Q.4(a) Find the eigenvalues and corresponding eigenvectors. *(6 Marks)*

- Characteristic equation – **2 Marks**
- Correct eigenvalues – **1 Mark**
- Eigenvector for first eigenvalue – **1.5 Marks**
- Eigenvector for remaining eigenvalue(s) – **1.5 Marks**

Q.4(b) Verify Cayley–Hamilton theorem and hence find A^{-1}. *(6 Marks)*

- Characteristic polynomial – **1.5 Marks**
- Verification of Cayley–Hamilton theorem – **2 Marks**
- Derivation of inverse using C-H theorem – **1.5 Marks**
- Final inverse matrix – **1 Mark**

OR

Q.5(a) Determine the modal matrix P and verify that $P^{-1}AP$ is diagonal. *(6 Marks)*

- Eigenvalues – **1 Mark**
- Eigenvectors – **2 Marks**
- Formation of modal matrix P – **1 Mark**
- Computation of $P^{-1}AP$ – **2 Marks**

Q.5(b) Reduce the quadratic form to canonical form by orthogonal transformation. *(6 Marks)*

- Formation of symmetric matrix – **1 Mark**
- Determination of eigenvalues/eigenvectors – **2 Marks**
- Construction of orthogonal matrix – **1 Mark**
- Canonical form – **2 Marks**

General Instructions to Examiners

- Award step-wise marks for every correct mathematical step.
- Deduct marks only for computational or conceptual errors.
- If the correct method is followed but arithmetic mistakes occur, award proportionate marks.
- Equivalent valid methods should be given full credit.
- Ensure that the total marks awarded do not exceed the maximum marks prescribed.